

Goodness-of-fit for multi-distribution neutrino cross-section measurements with shared events

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When multiple differential cross-section distributions are extracted from a common event sample with correctly evaluated inter-distribution correlations, the statistical covariance matrix is generically singular. Exact linear constraints from event sharing across measurements, such as equal distribution totals when all selected events populate bins in every distribution, produce directions with zero statistical variance. Systematic uncertainties restore formal invertibility but leave near-singular directions that yield unphysically inflated χ^2 values in global goodness-of-fit tests. We present a projection method that restricts the test to the subspace carrying independent statistical information, yielding a χ^2 distributed with $N_{\text{bins}} - N_{\text{null}}$ degrees of freedom, where N_{null} is the number of independent constraints. The method is connected to established results on hypothesis testing with singular covariance matrices and validated with both an analytical toy model and a realistic simulated neutrino–argon cross-section measurement including unfolding and parametrized multi-source systematic uncertainties.

I. INTRODUCTION

Modern neutrino cross-section literature is experiencing an increase of published measurements reporting multiple differential distributions extracted from the same event sample. MicroBooNE has reported simultaneous measurements of cross sections for final states with and without protons [1, 2], multi-differential QE-like cross section measurements in kinematic imbalance variables [3, 4], and triple-differential inclusive results [5]. Other relevant examples include the multiple final-state kinematic projections from T2K [6, 7], and double-differential inclusive results from NOvA [8–10]. In each of these analyses, the same selected events populate bins in multiple reported distributions, leading to strong inter-measurement statistical correlations. Separate but related efforts have produced simultaneous measurements across different nuclear targets [11–13] and combined neutrino-antineutrino results [14], where correlations arise from shared systematic uncertainties rather than shared events.

The blockwise unfolding framework [15] provides a rigorous method for computing the full cross-distribution covariance matrices in all such measurements. In this approach, all bins corresponding to the same measured observable in the multi-distribution analysis are organized in blocks, each corresponding to a single differential cross-section distribution. Each bin block is unfolded independently using any standard method (e.g. iterative D’Agostini unfolding [16] or Wiener-SVD [17]), while the full covariance matrix, encoding the correlations between bins from different distributions, is constructed and propagated through the cross section extraction procedure. The statistical covariance for bins in different blocks is determined by the number of events they share. The result is a single combined covariance matrix covering all reported bins across all distributions.

A natural use of these combined measurements is com-

puting global goodness-of-fit metrics, such as a χ^2 statistic, that test a model against all distributions simultaneously. Compared to evaluating each distribution independently, a global test that accounts for all inter-block correlations provides the maximum discriminating power available from the data. Such global metrics are essential for interaction model tuning efforts [18–20], where measurements of multiple kinematic distributions provide complementary sensitivity to different aspects of the underlying physics. However, as we show in this paper, the full inter-block statistical covariance is generically singular when it correctly accounts for correlations from shared events. The rank deficiency arises from exact linear constraints imposed by the event-sharing structure of the binning scheme: certain linear combinations of bins, such as the difference in total event counts between two distributions of the same events, have identically zero variance. While systematic uncertainties formally restore invertibility, the resulting near-singular directions produce unphysically inflated χ^2 values, yielding a global test with the wrong effective number of degrees of freedom.

We present a method based on projecting onto the range (column space) of the statistical covariance matrix that yields a well-defined chi-squared with the correct number of degrees of freedom. We demonstrate the method using a pedagogical toy example and connect it to the established statistical literature on hypothesis testing with singular covariance matrices.

Brief roadmap of sections.

II. ORIGIN OF THE SINGULARITY

Consider a neutrino cross-section analysis simultaneously measuring B different flux-integrated differential cross sections, each with N_b true bins. If the measurement is extracted from a common event sample then each

selected event will populate exactly one bin in each distribution it belongs to. Recent MicroBooNE analyses [21] have adopted the blockwise unfolding approach presented in Ref. [15] to simultaneous cross-section measurements.

This method provides a systematic framework for reporting the correlations between the different extracted differential cross sections. It provides a general prescription to construct statistical covariance matrices that respect the correlations between bins in different distributions or blocks. The full inter-block covariance matrix constructed in reconstructed space is propagated through unfolding using a block-diagonal unfolding matrix, obtained as a direct sum of the individual unfolding matrices computed for each block. The result of the combined measurement is a vector of $N_{\text{bins}} = \sum_{b=1}^B N_b$ cross-section values with an $N_{\text{bins}} \times N_{\text{bins}}$ covariance matrix.

A. Statistical covariance from shared events

The core idea behind the treatment of the statistical correlations in the blockwise unfolding approach is the partition of any arbitrary pair of overlapping kinematic bins X and Y into a set of three non-overlapping bins u , v , and w . The events in the intersection $X \cap Y$ are assigned to bin v , while bin u contains the events in the relative complement of X in Y (i.e. $X \setminus Y$), and bin w corresponds to the relative complement of Y in X (i.e. $Y \setminus X$). Denoting the number of predicted events in each bin as n_i , and assuming these follow independent Poisson distributions, the covariance between the counts in bins X and Y is given by [15]:

$$\begin{aligned} \text{Cov}(n_X, n_Y) &= \text{Cov}(n_u + n_v, n_v + n_w) \\ &= \text{Cov}(n_u, n_v) + \text{Cov}(n_u, n_w) \\ &\quad + \text{Cov}(n_v, n_v) + \text{Cov}(n_v, n_w) \\ &= \text{Var}(n_v), \end{aligned} \quad (1)$$

as by definition bins u , v , and w have no events in common. This implies that the covariance between any two bins can be estimated from the variance of the events they share.

The variance for a set of Monte Carlo (MC) events can be estimated from the sum of their squared weights. This gives us a general prescription to calculate the elements of the statistical covariance matrix as:

$$(C_{\text{stat}})_{ij} = \sum_{e \in (i \cap j)} w_e^2, \quad (2)$$

where w_e is the weight associated to event e belonging to the intersection of bins i and j .

After propagation through the block-diagonal unfolding, the statistical covariance in unfolded space inherits the same rank structure, given that the unfolding matrix is invertible. This rank structure has a direct interpretation in terms of the number of distinct event-sharing patterns across distributions.

B. Combination matrix and null space

For each event e , we can define the indicator vector $\phi_e := (\phi_i)_{i=0}^{N_{\text{bins}}}$ such that $\phi_i = 1$ if $i \in \text{bins}(e)$ and $\phi_i = 0$ if $i \notin \text{bins}(e)$, where $\text{bins}(e)$ is the set of bins event e populates. The statistical covariance can be re-written in terms of indicator vectors as:

$$C_{\text{stat}} = \sum_e w_e^2 \phi_e \phi_e^T, \quad (3)$$

where w_e is the weight associated to event e .

The rank of the statistical covariance matrix equals the dimension of $\text{span}\{\phi_e\}$, which depends only on the distinct indicator vectors present, not on their weights. Define the bin-combination matrix A as the matrix whose rows are the unique indicator vectors among ϕ_e . Then:

$$\text{rank}(C_{\text{stat}}) = \text{rank}(A) \quad (4)$$

A vector $\mathbf{x}$ is in the null space of C_{stat} if and only if, for every event e :

$$\sum x_i = 0 \quad (5)$$

C. Types of constraints

The number of null directions N_{null} depends on the overlap structure between blocks. The constraints described above hold exactly when every selected event populates at least one bin in every block, as is the case when underflow and overflow bins are included. Without such bins, events falling outside the kinematic range of a given block do not contribute to the inter-block sum rules, potentially reducing N_{null} . In practice, the inclusion of overflow bins is already recommended for proper unfolding closure [15], and any well-designed blockwise analysis will therefore exhibit the full set of constraints. The projection method presented here remains valid regardless of the value of N_{null} , including the limiting case $N_{\text{null}} = 0$ where C_{stat} is full rank and the projector reduces to the identity.

D. Effect of systematic uncertainties

E. Ignoring inter-block correlations

III. THE PROJECTION METHOD

A. Constructing the projector

B. Projected chi-squared

C. Proof sketch / justification

D. Relation to alternative approaches

IV. ANALYTICAL DEMONSTRATION

A. Setup

B. Eigenvalue analysis

C. Pseudo-experiment validation

D. Sensitivity to threshold choice

V. REALISTIC DEMONSTRATION WITH NEUTRINO GENERATORS

A. Simulation setup

B. Example 1: Two blocks ($N_{\text{null}} = 1$)

C. Example 2: Four blocks with multiplicity slicing ($N_{\text{null}} = 2$)

D. Practical observations

VI. PRACTICAL IMPLEMENTATION GUIDE

A. Step-by-step recipe

B. What to include in data releases

C. Compatibility notes

VII. CONNECTION TO THE STATISTICS LITERATURE

VIII. SUMMARY AND RECOMMENDATIONS

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